Final Project - Data Analytics & Visualization

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1. Introduction

- Briefly introduce the project topic and its significance.

The intention of this project is to show a brief statistical information and to provide visualizations of the five main country of origin of international asylum seekers and refugees. I selected six indicators of different conditions that may impact their decision to leave their country. I want to show the correlation between the different variables within the indicators for the year 2020. And using three models of regression to predict migrants decision.

- Provide an overview of the dataset selected and its data sources.

The analysis on this project relies on several datasets. For the Refugees and asylum seekers population, I make use of the United Nation High Commissioner for Refugees (UNHCR) statistics. To derive various variables I use data from the Our World in Data and Uppsala Conflict Data Program (UCDP)

2. Data

- Describe the selected real-world dataset and its relevance to the project.

The data from UNHCR is the main dataset showing refugees and asylum seekers population, years, their country of origin and country of asylum. The other datasets selected shows different indicators;

- Unemployment rate in %
- Conflicts describing them by year and their victims
- Global food supply screening provision in Kcal, protein and fat per day
- Homicides showing the homicide rate per 100,000 population for both sexes and all ages
- Happiness and life satisfaction measured by the Cantril ladder score - - Public spending on education as a share of GDP showed as a %

I consider all the variables above as the main reasons to decide to attempt the refugee and asylum seeker status. The main difference between those two types of population is the legal status. Asylum seekers is someone who is seeking international protection from dangers in their home country but whose claim for refugee status hadn't been determined legally.

- Provide a reference to the source of the dataset. You can find the dataset [here](INSERT LINK TO DATA SOURCE).

The below datasets and sources have been used;

- ✓ Refugees & Asylum-Seekers population - source: UNCHR - This is the main dataset
- ✓ Population
- ✓ Unemployment
- ✓ Conflicts
- ✓ Food Supply
- ✓ Homicides
- ✓ Happiness and Life Satisfaction
- ✓ Education expenditure

✓ Country Income class

3. Importing

- Explain the method used to import the data into Python.
    - Import data from a flat file (e.g., CSV, XLS, XLSX, TXT).
    - Retrieve data using online SQL, APIs, or web scraping.

Mainly the data was extracted from flat files with extension . CSV and XLS using Panda Data-frames importing pandas as pd and pd.read_CSV (filename.csv)

4. Data Preparation

- Discuss the steps taken to prepare the data for analysis.

I used the CRISP-DM process following the 6 phases;

Business Understanding

I used Refugees and Asylum seekers population distributed by genre data as fact table to understand the variables of their home countries in order to take the decision to pursue refugee status. With this factor table I also used another dimensional tables containing variables of their country of origin to predict their request of status, my first step was analysing all variables from all home countries to determine which ones are the top 5 examining only 2020 - as it's the year in which I had full information from al datasets.

Data understanding

I used several datasets and combined all of them.

- The main dataset is the forced migration data - refugees and asylum seekers. It shows their country of origin, country of asylum and genres distribution.
- Economic Indicator: The unemployment data which is a key determinant of origin and destination country shows the rate expressed in percentages.
- Conflict Indicator: This dataset shows location, estimation of fatalities and the event in which caused the number of fatalities.
- Food Indicator: Food supply shows population and the food supply in kilo-calories, proteins and fat grams per day.
- Violence Indicator: Homicides display the rate per 100,000 population in both sexes and all ages.
- Living condition Indicator: Happiness and life satisfaction dataset uses the Cantril Ladder Scale (CLS) which its a scale that assess general life satisfaction - the score top of the ladder represent the best possible life for a person and the bottom would be the worst - from 0 to 10.
- Education Indicator: Education spending shows estimates of expenditures from the government, which in most countries nowadays are perceived as a duty from the government to ensure public education.

All the above datasets contain country and years from 2012 to 2022 data.

Data Preparation

The dataset contains some missing values which I had to fix replacing the NaN with 0. Also I renamed a lot of column's name

In the first phase of the analysis, I used the whole data from all countries of origin and countries of asylum and genre distribution (female/male) to see some trends and gather summary statistics (mean, median, maximum, minimum and standard deviation) also I did the same for all indicators data sources
Then I subset the countries of origin in filtering the ones with the highest population which are five in total and performing all type of data manipulation
In the final stage I used country as the key ID to merge different sets of data and have all data into one DataFrame to see correlation of different variables and to separate the data into train data and test data to run the linear regression models.

Models

I used the following Regression models: Linear Regression, Ridge and Lasso (Least Absolute Shrinkage and Selection Operator Regression), the last two models are the regularized version of Linear Regression.

Evaluation

I used the Mean Absolute Error (MAE) the best is close to 0

Deployment

The output could be used for several purposes, for example, studies on how to foresee refugee movements

- Explain the creation of pandas DataFrames.

I open different websites to gather data and downloaded under the extensions of .CSV and XLSX, imported pandas and assigned the filename.
By calling read_csv() I created a dataFrame which is the main structure used in Pandas. Also I have I loaded .xls files and loaded using pd.read_excel
Then I performed data manipulation to all dataFrames - applying all attributes as .shape, .columns, .dtypes, .info to be able to see data types, number of columns, rows, calculating the standard deviation for all numeric columns

- Describe the sorting, indexing, filtering, and grouping operations performed on the data.

I performed different filtering of data to subset rows of countries with the top 5 countries of origin with highest refugee population, selecting a determined year also using ‘and’ & ‘or | operators to filter more than one condition.
I used .groupby to group different variables and calculate summaries statistics. This operation to have different views of groupings of data combining the results.
I did use index which allow me to designate the Country Column as an Index and subset using .loc the top 5 countries in each dataset

- Explain how duplicate entries and missing values were handled.

I had several missing values across all datasets in which I replaced with 0 using .fillna(0) and after merging the main dataset (refugees population) with the others I had duplications so I had to subset the main dataset before merging with the others

- Discuss the definition of custom functions for reusable code.

Defining a function and then call it is very useful for the fact that it saves time in not writing it every time that you need to use it. Unfortunately I didn't use any function in my code.

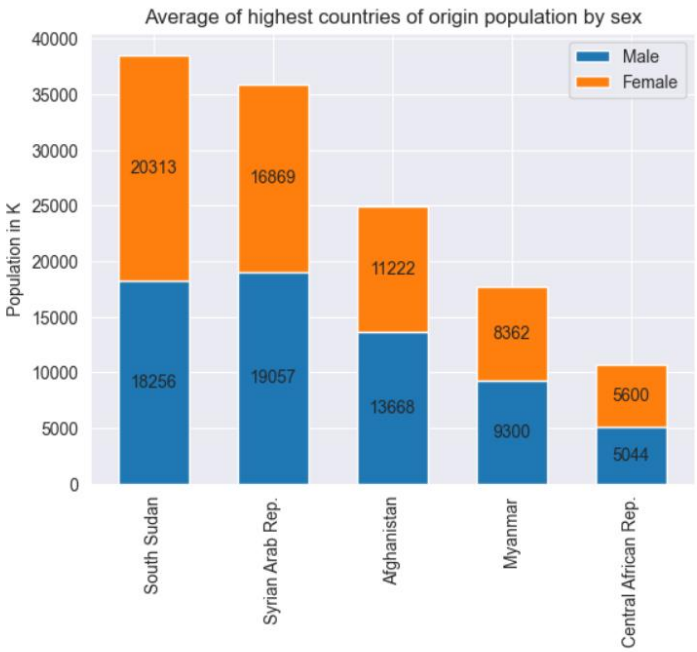
- Provide details on how multiple DataFrames were merged, if applicable.

As my analysis contain insights for five countries and six indicators that could affect the population's decision in requesting refugee status. To reach this analysis I did several merging for all datasets - first unemployment and food supply data sources into one, then I merged this one with homicides data, then I merged with happiness and then with education. Then I transformed the main data source in dropping all column and leaving only country, year and total of population after that I merged this with the latest merging including education

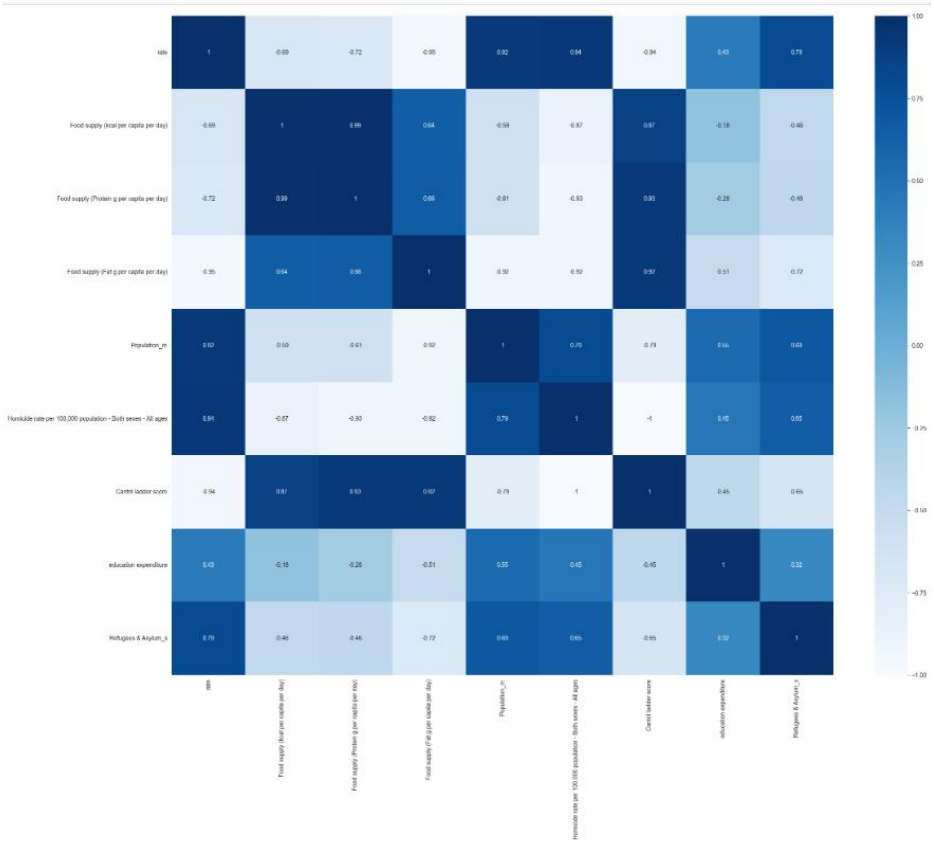
5. Data Visualization

- Generate at least four charts using the Matplotlib library

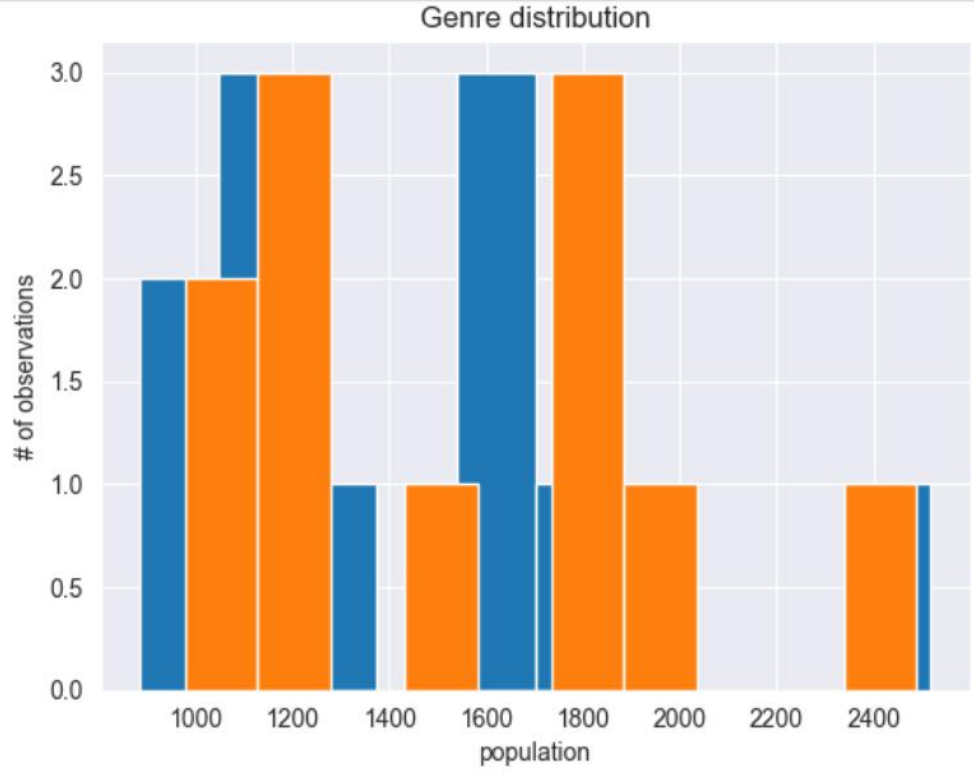
In order to show the distribution of females and males per country I used a stacked columns visualization from Matplotlib library



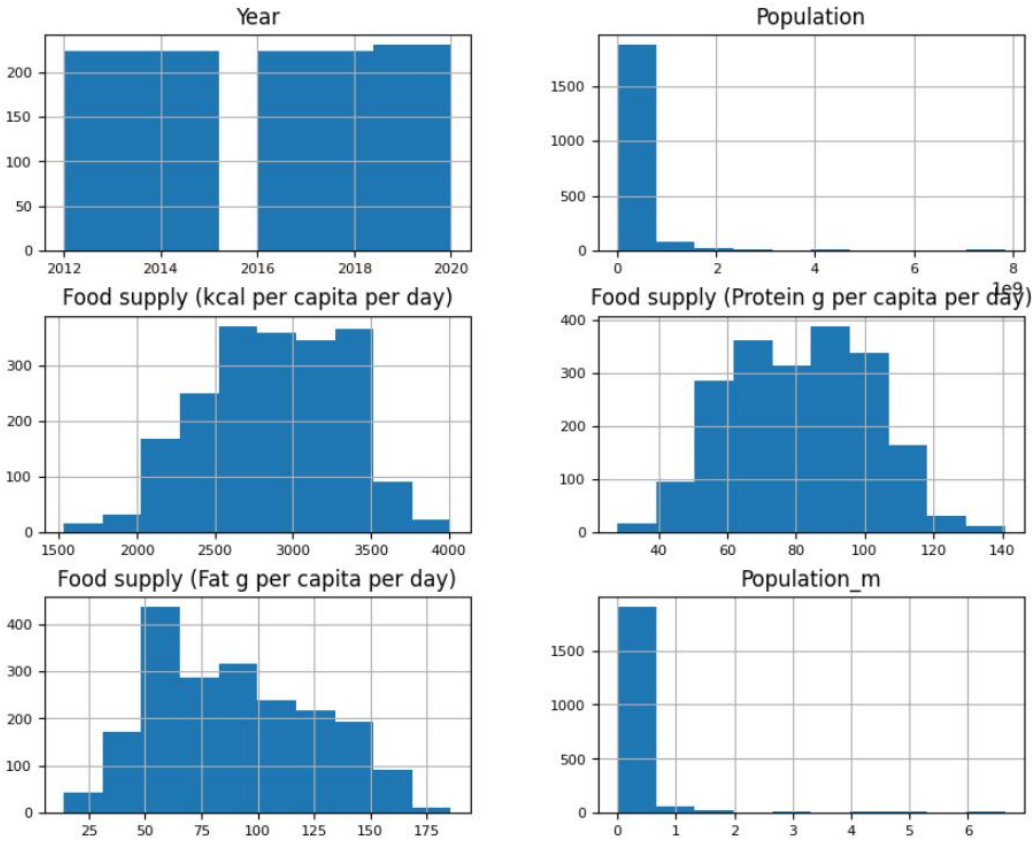
This graph shows the correlation of all variables from all datas sources



Showing the distribution of genres within the population of migrants

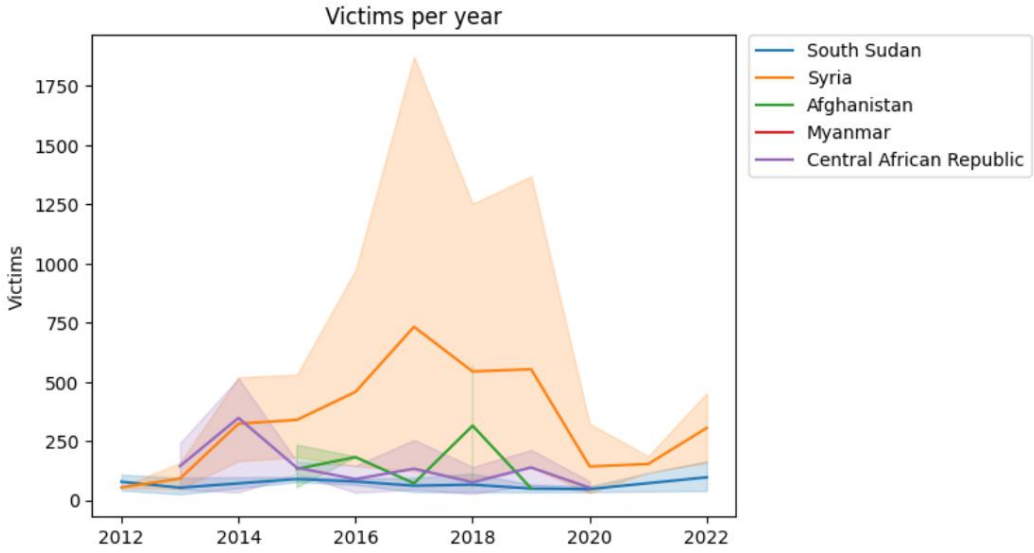


Showing an histograms of all numerical variables within food supply

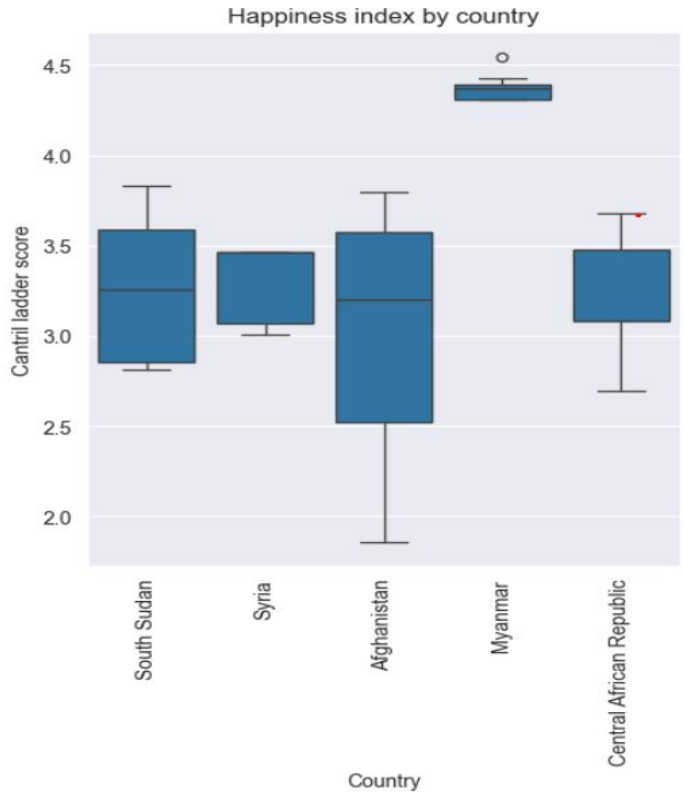


- Generate at least four charts using the Seaborn library.

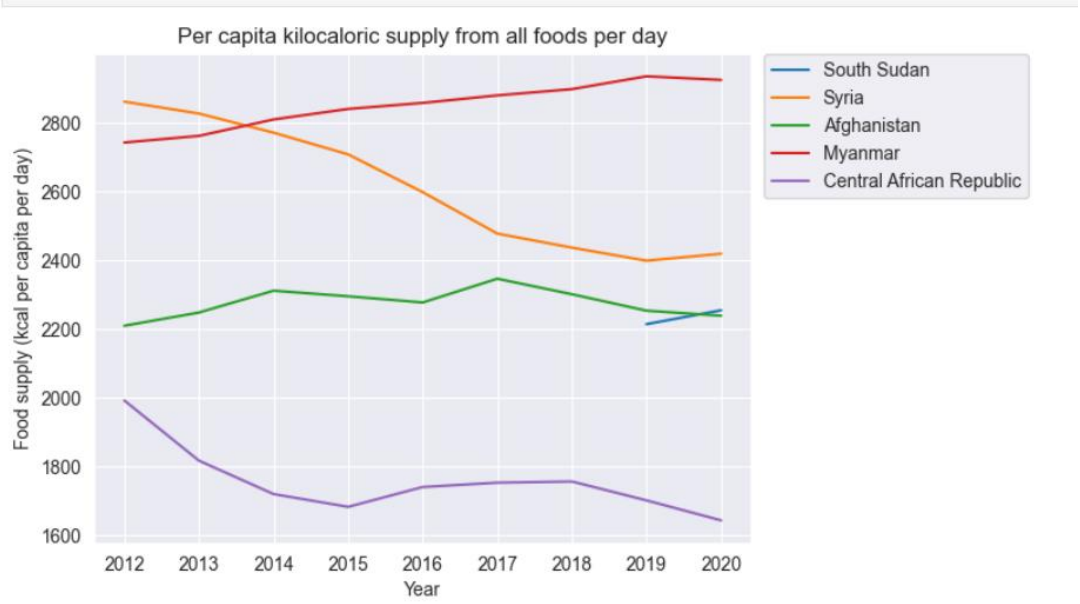
I used the below line plot from Seaborn library to show the relation between the victims of the 5 countries selected and trend by year showing the standard deviation



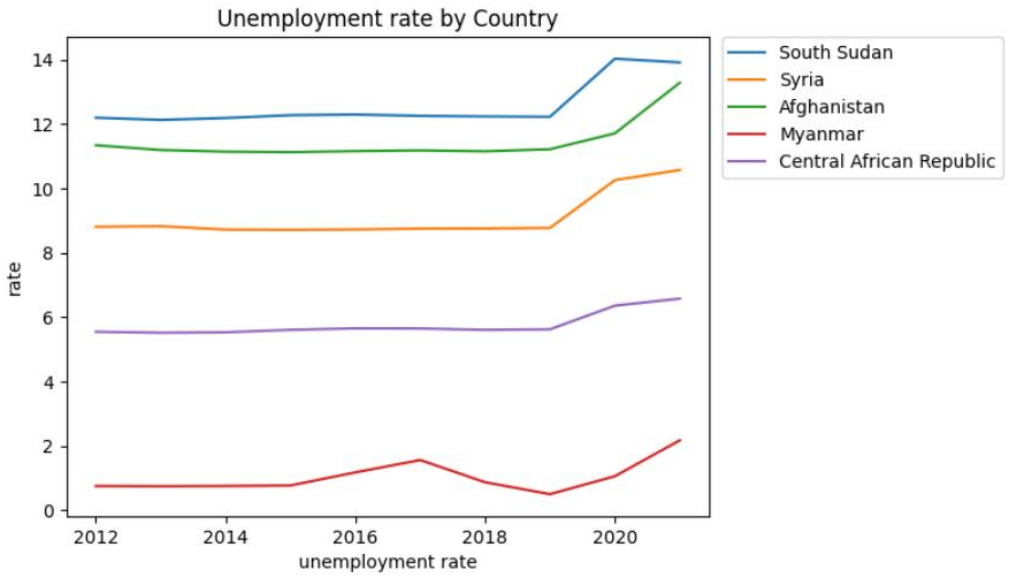
And to show life standard & happiness index I selected a catplot to show distribution of life index of the selected countries, also we can see some statistics as the median, percentiles and whiskers screening the spread distribution and an outlier



Below, we can see a lineplot showing the trend of the kilo-caloric supply food per day.

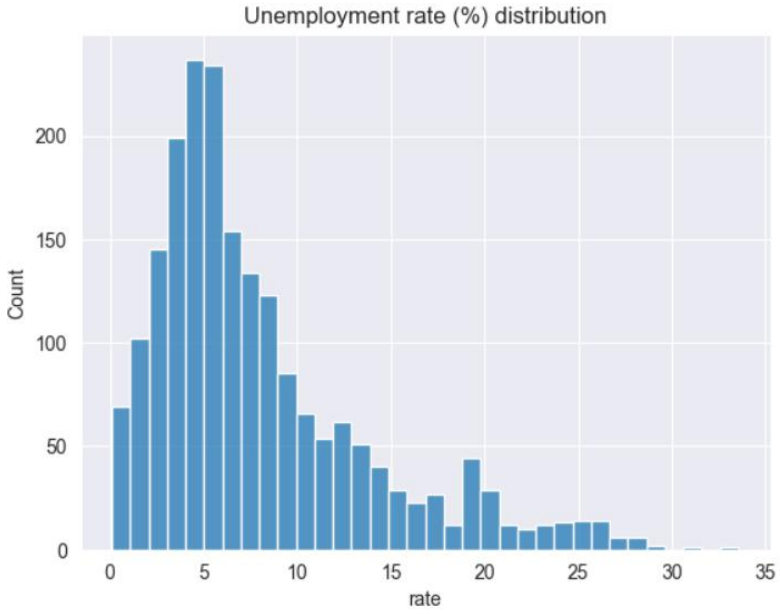


And below in this graph I'm showing the unemployment rate (%) trend by country

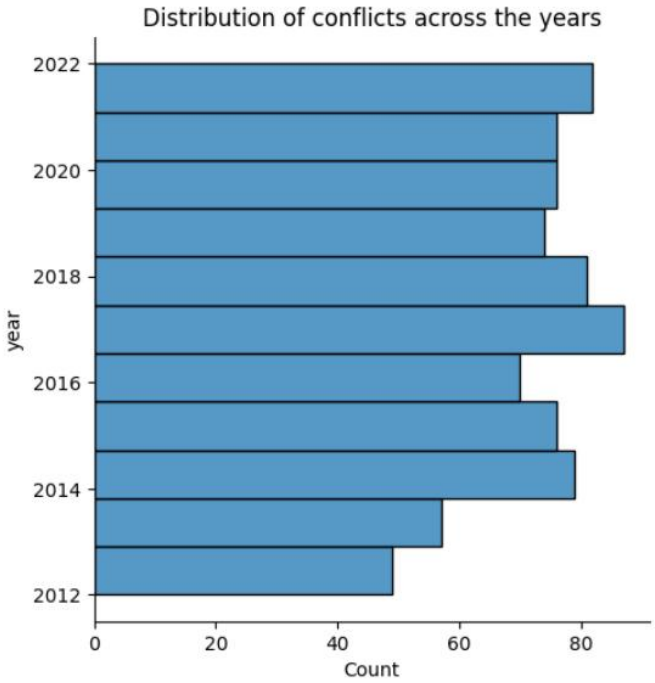


- Conduct univariate and bivariate analysis using appropriate charts and techniques.

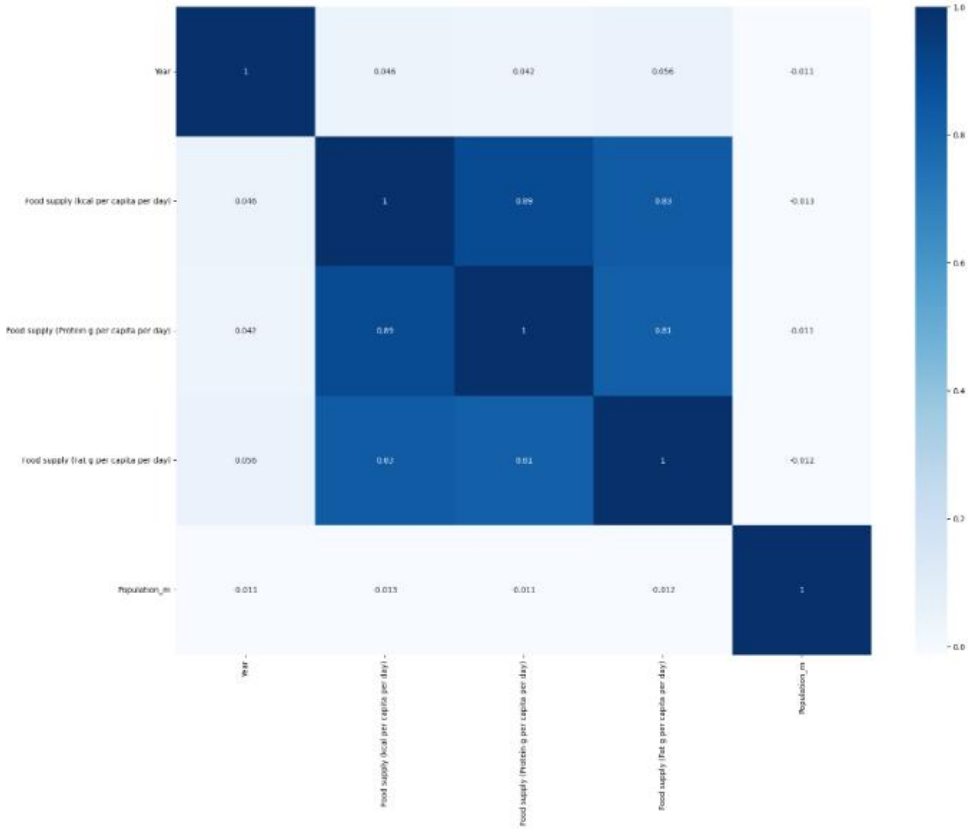
Univariates analysis. only one variable help to understand distribution, central tendency. I included some histograms as the below that shows the unemployment %rate distribution

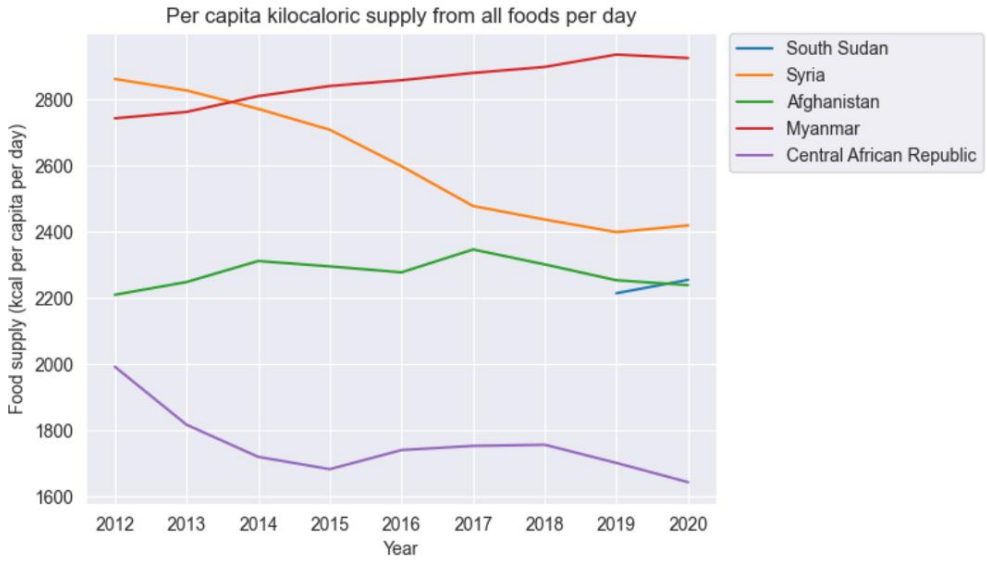


And distribution of conflicts across the years



Bivariate analysis: shows relationships between two variables, uncover associations, dependencies or correlation between variables. I included a heat-map and line plot graphs





6. Insights

- Derive at least eight valuable insights from your data analysis.
- Justify each insight with reference to the charts or analysis performed.

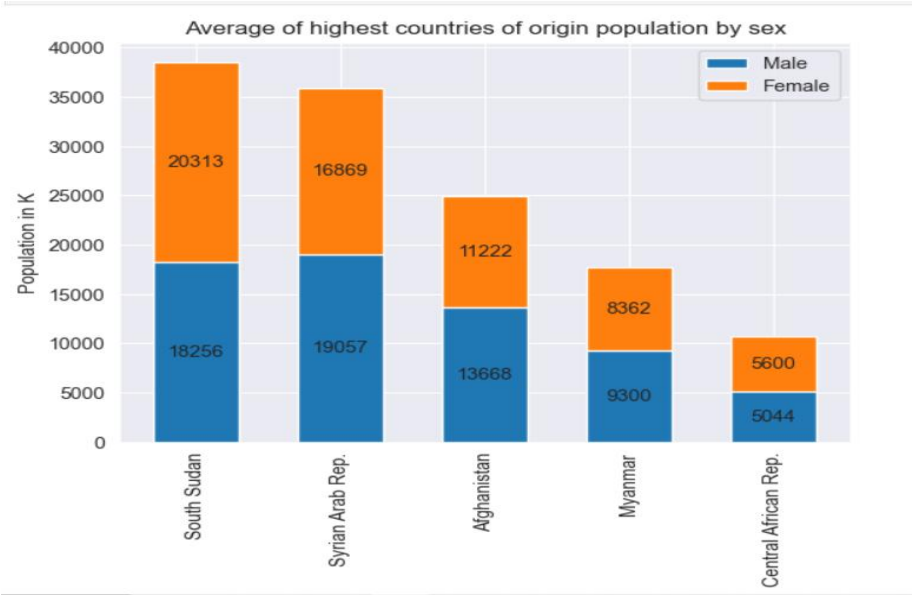
1- The majority of Females refugees/Asylum seekers come from South Sudan and majority of Males come from Syria

```
print((female_o).round(1))
```

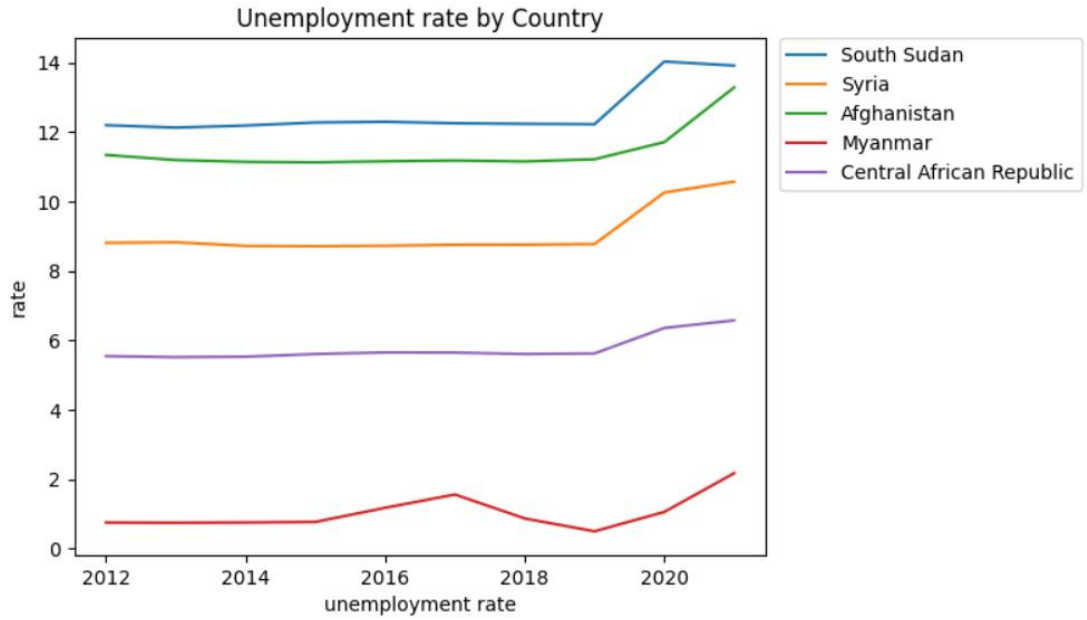
```
Country
South Sudan                  20312.9
Syria                        16869.2
Afghanistan                  11221.7
Myanmar                       8361.9
Central African Republic      5599.6
Name: Female total, dtype: float64
```

```
print((male_o).round(1))
```

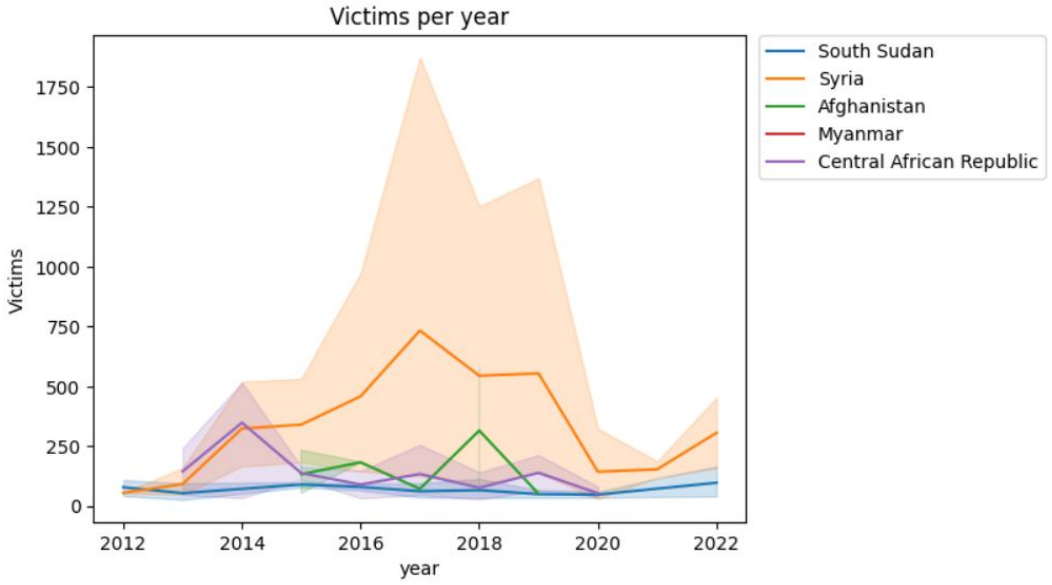
```
Country
Syria                        19056.9
South Sudan                  18255.8
Afghanistan                  13667.9
Myanmar                       9300.3
Central African Republic      5044.4
Name: Male total, dtype: float64
```



2 - South Sudan have the higher unemployment rate % with an average of 12% with the highest rate in 2020 of 14% followed by Afghanistan, which is the 3rd with the highest refugees.

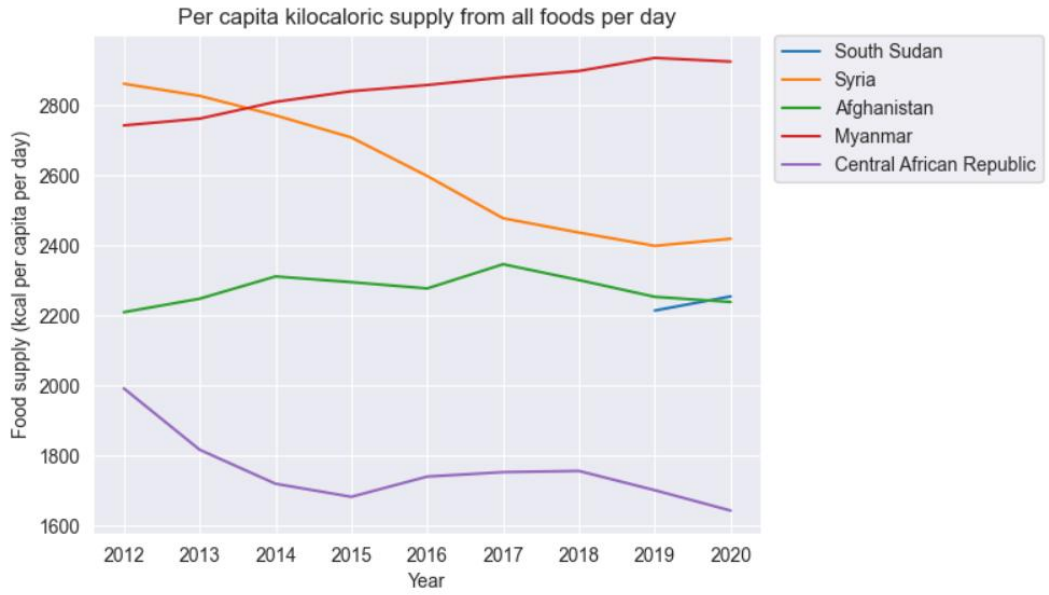


3. Syria have the highest number of victims in 2017 due to events involving IS

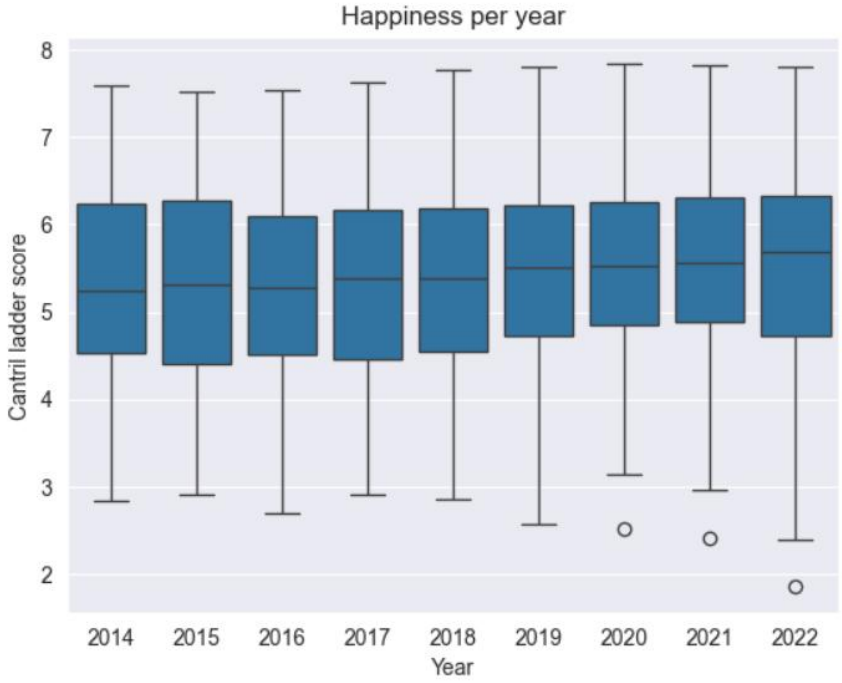


Country	side_a_name	year	Victims
Syria	IS	2017	9454
Syria	IS	2017	591

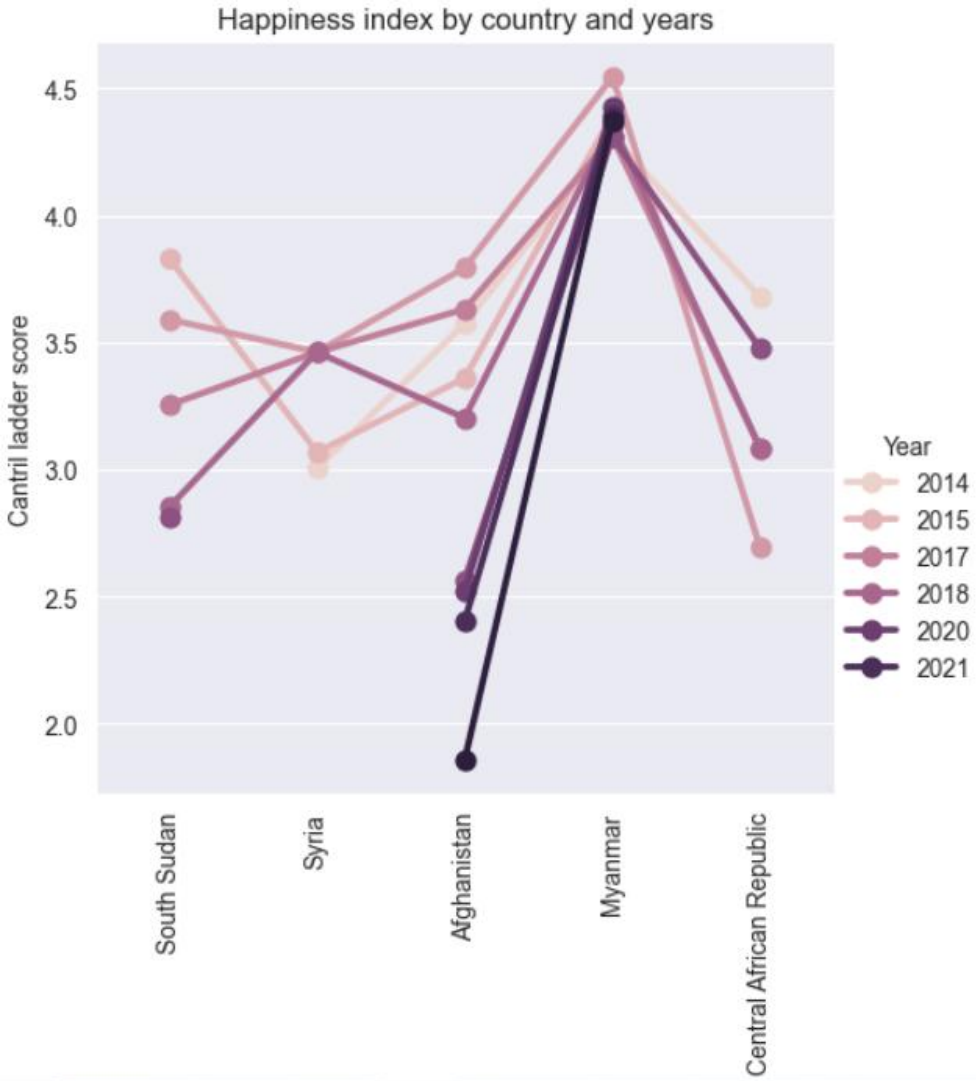
4. Myanmar is the country with the highest kcal per day which increased steadily over the years but 2012 Syria was in a better position than Myanmar but started to decline inn 2013 until 2020 which is the last year of data availability



5. In relation to happiness indicator, in general increased. As per the below box-plot we can see that the median increased nearly approaching the number 6 in scale in 2022 while in 2014 was close to 5. We have some outliers in 2020,2021 and 2022



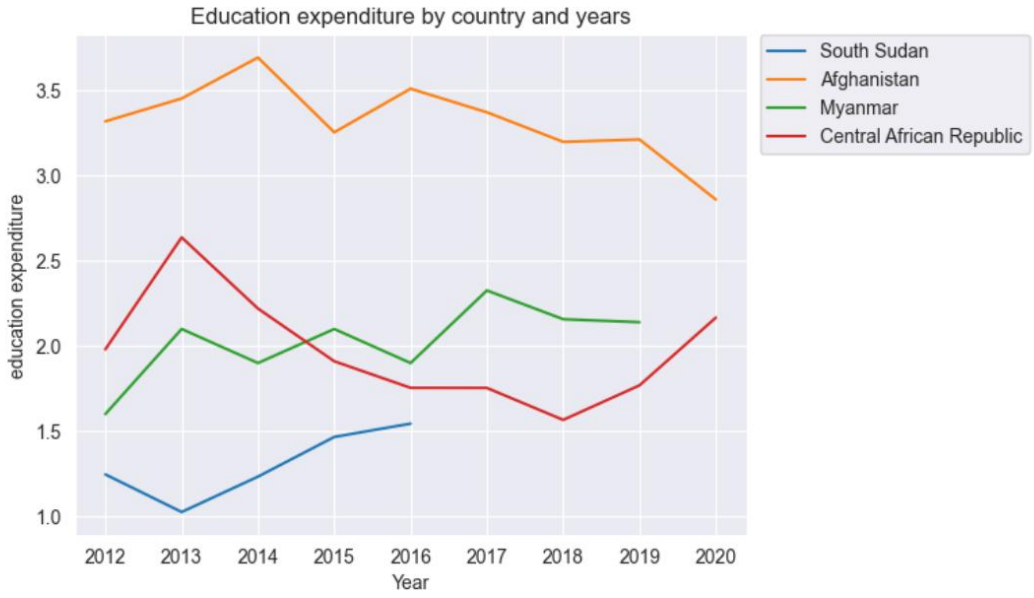
And as per below within the 5 countries in scope by far Myanmar is the happiest country, reaching nearly 4.5 in the Cantril scale. All the remaining have the same scale of 3.5



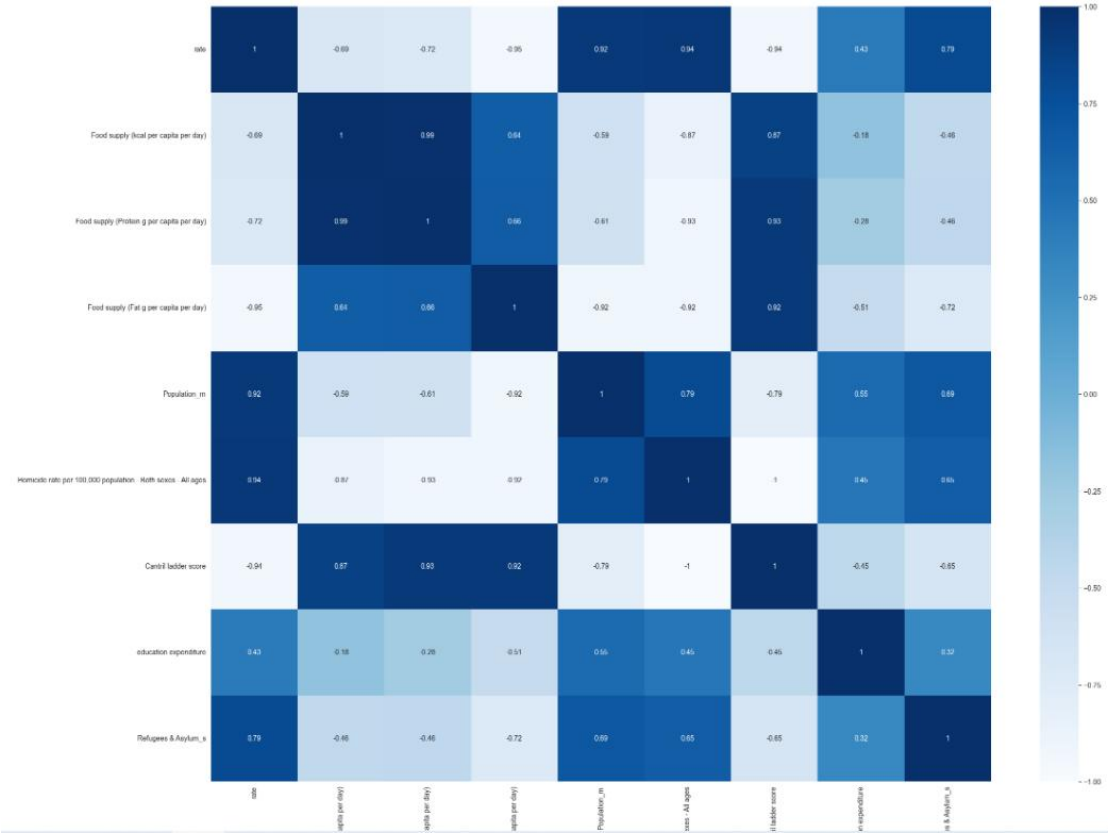
6. High number of homicides happened in Myanmar in 2021 by 28.43 per 100,000 population and seconded by South Sudan in 2012 by 14.05.

Country	Code	Year	Homicide rate per 100,000 population - Both sexes - All ages
Myanmar	MMR	2021	28.437815
South Sudan	SSD	2012	14.053967

7. Afghanistan in 2012 was the country with the highest education spending from all the sample reaching the expenditure peak in 2014, which as well was the year where it started to declined, even tough increased a little bit in 2015, it fall badly on the subsequent years.
Spending in Central Africa had a peak in 2013 but decreasing later but from 2018 is showing an upward trend.



8. As per the below heat-map - We have a high correlation between variables of different indicators, the higher one is Unemployment rate with homicides of 0.94.
Food supply (protein g per capita) and happiness index with a correlation of 0.93, a little bit higher than the fat g and happiness which is 0.92.
Another higher correlation is unemployment rate and Population of 0.92



9. As per the three regression models performed - The linear regression shows a coefficient of determination of 0.82(test accuracy) which is closer to 1 therefore we have a linear relationship. One of the problem that I had is that the scores in the below models were negative which I solved increasing the size of the data for training and testing

	Models	Scores
0	LinearRegression	0.829168
1	Ridge	0.817231
2	Lasso	0.345365

7. Machine Learning

- Describe potential predictions that could be made using machine learning and/or deep learning in the future.

Machine learning and deep learning have a great future ahead as both of them are very important to make predictions for example in the weather to foresee natural catastrophes and automatize actions

- Justify whether classification or regression methods would be more suitable for the identified predictions.

Both of them are within a supervised learning category, I consider that a regression method is more suitable to identify predictions as only needs a set of features called predictors to perform the task